

Quantum Query Algorithms for the Constructive Diagonal Ramsey Theorem

Anonymous author

Anonymous affiliation

Abstract

The constructive diagonal Ramsey problem asks for a clique or an independent set of the size guaranteed by Ramsey’s theorem, using adjacency queries to the input graph. Given coherent adjacency-oracle access to an N -vertex simple graph, we give a bounded-error quantum algorithm that, for every $K \geq 2$ and $N \geq 4^{K-1}$, finds and verifies a homogeneous K -set using

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

edge queries with failure probability at most η . At the Ramsey scale $N = 2^n$, the algorithm finds a homogeneous set of order $\lfloor n/2 \rfloor + 1$ using $O(\sqrt{N} \log N \log(\log N/\eta))$ queries, improving on the $O(N)$ query complexity of the standard explicit Erdős–Szekeres recursion.

The algorithm runs the constructive Ramsey recursion over implicit candidate sets. Each set is represented by a short conjunction of adjacency constraints and sampled using capped unknown-solution quantum search. A scale-aware concentration schedule then balances estimation accuracy against the increasing cost of sampling deeper sets. We also give an estimation-free size-biased recursion, extend it to every fixed number of edge colours, and prove an $\Omega(M^{1-1/\sqrt{2}})$ randomized classical lower bound in the regime $M = 4^{K-1}$.

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1 Introduction

Ramsey’s theorem guarantees that every sufficiently large graph contains a large clique or independent set. Its classical constructive proof [6] repeatedly chooses a pivot and restricts the candidate set to one of its two colour neighbourhoods. Starting from $M = 4^{K-1}$ vertices, truncating each retained neighbourhood to half its current size gives an $O(M)$ -query algorithm.

The central observation of this paper is that the candidate sets need not be materialized. After i rounds, membership in the current set is determined by the i adjacency constraints imposed by the previous pivots. These short predicates allow quantum search to access the same recursion at substantially smaller query cost. The threshold 4^{K-1} matches this recursive construction.

We work in the promised valid-graph adjacency-oracle model. One coherent query returns the colour of a requested unordered vertex pair, and the output is a set of

2 Quantum Constructive Ramsey Search

vertices inducing one colour. Our main result is the following.

► **Theorem 1.1** (Implicit-majority quantum Ramsey search). *Let $K \geq 2$, $N \geq 4^{K-1}$, and $0 < \eta < 1/2$. Given coherent edge-oracle access to a simple undirected graph on N vertices, there is a quantum algorithm which, with probability at least $1 - \eta$, outputs a K -clique or a K -vertex independent set using*

$$O\left(2^K K \log \frac{K}{\eta}\right)$$

edge queries in the worst case. The algorithm verifies every returned witness and may output $\perp$ on the exceptional event.

► **Corollary 1.2.** *For a promised simple graph on $N = 2^n$ vertices, one can find a clique or independent set of order $\lfloor n/2 \rfloor + 1$ using*

$$O\left(\sqrt{N} \log N \log \frac{\log N}{\eta}\right)$$

edge queries.

Two implicit-set recursions.

The first algorithm avoids majority estimation altogether. At each round it samples a uniform candidate and follows the colour of its edge to the current pivot. The selected colour class is therefore chosen with probability proportional to its size. A survival lemma shows that this size-biased recursion completes with probability greater than $1/2$ from 4^{K-1} vertices, yielding an $O(2^K K^2 \log K \log(1/\eta))$ -query algorithm.

The sharper algorithm estimates a majority colour at each pivot. Sampling a deep candidate set is exponentially more expensive than sampling an early one, so a uniform accuracy requirement wastes queries at the deepest levels. We instead allocate a fixed total error budget across the recursion: early majorities receive higher accuracy, while later majorities receive larger allowable error. A dyadic schedule balances these costs and gives Theorem 1.1. The size-biased argument also extends to every fixed number of edge colours; see Corollary C.1.

Contributions.

The paper makes three main contributions.

- It gives a worst-case $O(2^K K \log(K/\eta))$ quantum-query algorithm for constructive diagonal Ramsey search, or $\tilde{O}(\sqrt{N})$ queries at the succinct Ramsey scale.
- It develops two methods for navigating the implicit recursion: a size-biased rule with a direct survival analysis, and a scale-aware majority estimator whose accuracy budget is matched to the cost of each level.
- It extends the size-biased method to a fixed number of colours and proves a randomized classical lower bound of $\Omega(M^{1-1/\sqrt{2}})$ for $M = 4^{K-1}$.

74 A reported lower-bound discrepancy.

75 At the default $N = 2^n$, $K = n/2$ parameters, [10, Section III, p. 415] reports an
 76 $N^{1-o(1)}$ quantum-query lower bound for the same Ramsey relation. This creates an
 77 apparent discrepancy with Corollary 1.2. Section 6 aligns the two formulations and
 78 shows that direct substitution into the cited ingredients yields an exponent at most
 79 $1/24$, rather than $1 - o(1)$.

80 The rest of the paper defines the oracle model and capped sampler, presents the
 81 two recursions, proves their correctness and query bounds, derives the randomized
 82 classical lower bound, and compares the result with prior work.

83 2 Oracle model and capped uniform search

84 Let $G: \binom{[N]}{2} \rightarrow \{0, 1\}$ be a promised simple undirected graph, extended symmetrically
 85 to ordered pairs with $G(u, u) = 0$. Its coherent edge oracle acts as

$$86 \quad O_G|u, v, b\rangle = |u, v, b \oplus G(u, v)\rangle.$$

87 Query complexity counts calls to O_G . Reversible operations on vertex labels are free, as
 88 usual in the query model, and coherent access to O_G is part of the input specification.

89 The same asymptotic bound applies to the standard locally verifiable TFNP relation
 90 with arbitrary Boolean-circuit encodings. Given a circuit $C(u, v)$, define the canonical
 91 simple graph

$$92 \quad A(u, v) = \begin{cases} C(\min\{u, v\}, \max\{u, v\}), & u \neq v, \\ 0, & u = v. \end{cases}$$

93 One query to A uses $O(1)$ queries to C , including uncomputation. Run Theorem 1.1
 94 on A and let W be the returned homogeneous set. Query the $O(K^2)$ diagonal and both
 95 ordered entries induced by W . A self-loop or an asymmetric pair is a local invalidity
 96 certificate; otherwise C agrees with A on W , so W is a valid clique or independent-set
 97 witness.

98 ► **Corollary 2.1** (Locally verifiable arbitrary encoding). *The standard Ramsey TFNP*
 99 *relation that accepts either a local invalidity certificate or a locally verified homogeneous*
 100 *set has the same $O(2^K K \log(K/\eta))$ quantum-query upper bound.*

101 Our algorithms use the following consequence of unknown-solution quantum search
 102 [2]. We include a detailed formulation in Appendix A because the analysis needs both
 103 bounded termination and exact conditional uniformity.

104 ► **Lemma 2.2** (Capped uniform marked-item search). *Let $T \subseteq [M]$ have density at*
 105 *least $\lambda > 0$, and suppose its membership predicate costs q edge queries. There is a*
 106 *quantum block which uses at most $O(q/\sqrt{\lambda})$ edge queries, succeeds with probability*
 107 *at least an absolute constant $p_0 > 0$, and, conditional on verified success, returns an*
 108 *exactly uniform element of T . The same predetermined cap makes the block terminate*
 109 *if the density promise is false or T is empty.*

110 Moreover, m independent uniform samples can be collected, except with probability
 111 δ , using a predetermined $O(m + \log(1/\delta))$ blocks whenever the density promise holds.

The batch guarantee avoids a separate $\log(1/\delta)$ overhead for each sample. Every block starts from the uniform state, uses fresh internal randomness, measures a candidate, and verifies membership. Conditional on success, the returned vertex is uniform on T ; successful outputs from distinct blocks are independent.

For comparison, consider the explicit classical recursion on $M = 4^{K-1} = 2^{r+1}$ vertices, where $r = 2K - 3$. At level i , query all edges from a pivot to the other $s_i = M/2^i$ candidates. A majority colour contains at least $s_i/2$ of the $s_i - 1$ non-pivot vertices; retain exactly $s_i/2$ of them. Then

$$\sum_{i < r} (s_i - 1) < 2M,$$

and two vertices remain after r rounds. The quantum algorithms retain the entire selected colour class but access it only through its membership predicate.

3 Two implicit-set recursions

3.1 An estimation-free size-biased recursion

Set $M = 4^{K-1}$ and $r = 2K - 3$, restrict the input to its first M vertices, let $S_0 = [M]$, and fix any $v_0 \in S_0$. At step $i = 0, \dots, r-1$, define $T_i = S_i \setminus \{v_i\}$. Apply the repeated capped search from Lemma 2.2, with the universal density lower bound $1/M$, to sample an exactly uniform $x_i \in T_i$; abort if the fixed cap is exhausted. Query

$$c_i = G(v_i, x_i),$$

and set

$$S_{i+1} = \{x \in T_i : G(v_i, x) = c_i\}, \quad v_{i+1} = x_i.$$

After the final step, put $w = v_r$. Among the r labels c_i , choose a colour that appears at least $K - 1$ times. Return the corresponding pivots together with w , after directly verifying the induced edges.

The algorithm never materializes S_i . Its membership predicate is the conjunction of the edge constraints imposed by the earlier pivots, together with the exclusions of those pivots, and costs $O(i+1)$ edge queries. Repeating each capped block $O(\log(2r))$ times reduces its conditional failure probability on a nonempty set to $O(1/r)$. The probability that all r implicit sets remain nonempty is analyzed in Lemma 4.1.

3.2 Scale-aware approximate majorities

The sharper recursion estimates a near-majority colour at each pivot. Fix $M = 4^{K-1}$ input vertices and set

$$r = 2K - 3, \quad E = \frac{1}{16}.$$

For $0 \leq i < r$, define

$$d_i = \left\lceil \frac{r-1-i}{6} \right\rceil, \quad z_i = 2^{-d_i}, \quad Z = \sum_{j=0}^{r-1} z_j,$$

146 and allocate

$$147 \quad \varepsilon_i = E \frac{z_i}{Z}, \quad a_i = \frac{1}{2} - \varepsilon_i. \quad (1)$$

148 Thus $\sum_i \varepsilon_i = E$. The allowable error doubles once every six levels, matching the
149 increasing cost of sampling the deeper candidate sets.

150 Before round i , the recorded pivots $v_0, \dots, v_{i-1}$ and colours $c_0, \dots, c_{i-1}$ define the
151 exact implicit set

$$152 \quad S_i = \{u \in [M] \setminus \{v_0, \dots, v_{i-1}\} : G(v_j, u) = c_j \text{ for every } j < i\}. \quad (2)$$

153 Its reversible membership predicate uses $O(i)$ edge queries.

154 At round i , use Lemma 2.2 to obtain a verified pivot $v_i \in S_i$. Then collect, with
155 replacement,

$$156 \quad m_i = \left\lceil \frac{1}{2\varepsilon_i^2} \log \frac{4r}{\eta} \right\rceil \quad (3)$$

157 independent uniform samples from $S_i \setminus \{v_i\}$. Query their edges to v_i , and let $\hat{p}_i$ be
158 the observed fraction of colour 1. Set $c_i = 1$ when $\hat{p}_i \geq 1/2$ and $c_i = 0$ otherwise.
159 Appending this edge constraint to Equation (2) defines S_{i+1} .

160 After r rounds, one final capped search returns $w \in S_r$. Select a colour that occurs
161 at least $K - 1$ times among $c_0, \dots, c_{r-1}$, take any $K - 1$ pivots carrying that colour,
162 and add w . Query all $\binom{K}{2}$ induced edges and return the set only if this direct check
163 confirms homogeneity.

164 Every search and sampling batch has a predetermined cap. Hence every execution
165 terminates even if an inaccurate earlier estimate invalidates a later density promise. A
166 failed search or batch returns $\perp$.

167 **4 Correctness and survival of the nested sets**

168 We first analyze the estimation-free recursion. For integers $d, s \geq 0$, let $P_d(s)$ be the
169 infimum, over all graphs, candidate sets of size s , and distinguished pivots in those
170 sets, of the probability that the ideal size-biased recursion completes d additional steps.
171 Set $P_0(s) = 1$ for $s \geq 1$, $P_d(0) = 0$, and $P_d(1) = 0$ for $d \geq 1$.

172 ► **Lemma 4.1** (Size-biased survival). *For every $d \geq 0$ and $s \geq 1$,*

$$173 \quad P_d(s) \geq \frac{(s - 2^d + 1)_+}{s}, \quad (y)_+ = \max\{y, 0\}.$$

174 *In particular, for $M = 4^{K-1}$ and $r = 2K - 3$, the ideal recursion completes with*
175 *probability at least $1/2 + 1/M$.*

176 **Proof.** The case $d = 0$ is immediate. Suppose $d \geq 1$ and $s \geq 2$, and let the two colour
177 classes among the $s - 1$ non-pivot vertices have sizes a and b . A uniform sample selects
178 these classes with probabilities $a/(s - 1)$ and $b/(s - 1)$. The induction hypothesis
179 therefore gives

$$180 \quad P_d(s) \geq \frac{aP_{d-1}(a) + bP_{d-1}(b)}{s - 1} \geq \frac{(a - 2^{d-1} + 1)_+ + (b - 2^{d-1} + 1)_+}{s - 1}.$$

181 Because $(x)_+ + (y)_+ \geq (x+y)_+$ and $a+b = s-1$, the numerator is at least $(s-2^d+1)_+$.
 182 Replacing the denominator $s-1$ by s weakens the bound and proves the claim; the
 183 definitions cover $s \leq 1$. Finally, $M = 2^{r+1}$, so the bound at $(d, s) = (r, M)$ is
 184 $(M - 2^r + 1)/M = 1/2 + 1/M$. ◀

185 Exact conditional uniformity of each successful capped search couples an imple-
 186 mented run to the ideal recursion. Assign each of the r searches failure probability
 187 at most $1/(100r)$. A union bound decreases the completion probability by at most
 188 $1/100$, so a single fixed-cap run succeeds with probability at least 0.49. Independent
 189 repetition, followed by exact verification, amplifies this probability to $1 - \eta$ using
 190 $O(\log(1/\eta))$ runs. On every completed run, nesting gives $G(v_i, v_j) = G(v_i, w) = c_i$ for
 191 $j > i$; the pigeonhole argument in Lemma 4.4 then gives a homogeneous output.

192 We next analyze the scale-aware recursion. The first lemma controls all adaptive
 193 estimates with one event.

194 ► **Lemma 4.2** (Conditional near-majorities). *Except with probability at most $\eta/2$, every*
 195 *completed sampling batch reached after successful searches and an accurate prefix*
 196 *satisfies*

$$197 \quad |\hat{p}_i - p_i| \leq \varepsilon_i,$$

198 where p_i is the true colour-1 fraction from v_i to $S_i \setminus \{v_i\}$.

199 **Proof.** Condition on the complete classical history $\mathcal{H}_i$ before the samples at level
 200 i . The graph, S_i , and v_i are then fixed. Let $\mathcal{B}_i$ be the event that the verified batch
 201 succeeds. Conditional on $\mathcal{B}_i$, Lemma 2.2 gives independent uniform samples from the
 202 fixed set. Hoeffding's inequality and Equation (3) imply

$$203 \quad \Pr(|\hat{p}_i - p_i| > \varepsilon_i \mid \mathcal{H}_i, \mathcal{B}_i) \leq 2e^{-2m_i\varepsilon_i^2} \leq \frac{\eta}{2r}.$$

204 Multiplying by $\Pr(\mathcal{B}_i \mid \mathcal{H}_i) \leq 1$ gives the same upper bound for a completed but
 205 inaccurate batch conditional on $\mathcal{H}_i$. Apply this bound to the first inaccurate completed
 206 level after an accurate prefix. The tower property removes the conditioning, and a
 207 union bound over the r possible levels completes the proof. The argument does not
 208 require independence across levels. ◀

209 There are r pivot searches, r sampling batches, and one final search. Give each
 210 operation failure probability at most $\eta/(4r+2)$ using the capped amplification in
 211 Lemma 2.2. Applying these conditional bounds after good prefixes, as above, shows
 212 that their union has probability at most $\eta/2$. Together with Lemma 4.2, all searches
 213 and estimates used below are good with probability at least $1 - \eta$.

214 Let $s_i = |S_i|$ and

$$215 \quad A_0 = 1, \quad A_i = \prod_{j=0}^{i-1} a_j.$$

216 ► **Lemma 4.3** (Variable recurrence and density). *On every prefix satisfying Lemma 4.2,*

$$217 \quad s_i > MA_i - 1, \quad A_i \geq \frac{7}{8} 2^{-i}.$$

218 *Consequently S_r is nonempty, and for every sampling level $i < r$,*

$$219 \quad \frac{|S_i \setminus \{v_i\}|}{M} > \frac{3}{8} 2^{-i}.$$

220 *The final set also satisfies $|S_r|/M \geq 2^{-(r+1)} \geq (3/8)2^{-r}$.*

221 **Proof.** An accurate empirical majority retains at least the fraction $a_i = 1/2 - \varepsilon_i$ of
222 the non-pivot vertices. Hence

$$223 \quad s_{i+1} \geq a_i(s_i - 1). \tag{4}$$

224 Since $\sum_i \varepsilon_i = 1/16$ and $\prod_j (1 - x_j) \geq 1 - \sum_j x_j$ for $x_j \in [0, 1]$,

$$225 \quad A_i = 2^{-i} \prod_{j < i} (1 - 2\varepsilon_j) \geq 2^{-i} \left(1 - 2 \sum_{j < i} \varepsilon_j \right) \geq \frac{7}{8} 2^{-i}.$$

226 We prove $s_i > MA_i - 1$ by induction. The claim holds at $i = 0$. If it holds at level
227 i , then Equation (4) and $a_i < 1/2$ give

$$228 \quad s_{i+1} > a_i(MA_i - 2) = MA_{i+1} - 2a_i > MA_{i+1} - 1.$$

229 Because $M2^{-r} = 2$, we obtain $s_r > (7/8)2 - 1 = 3/4$, and therefore the integer s_r is
230 positive. For $i < r$, we have $M2^{-i} \geq 4$ and

$$231 \quad s_i - 1 > MA_i - 2 \geq \frac{7}{8} M2^{-i} - 2 \geq \frac{3}{8} M2^{-i}.$$

232 Finally, $s_r \geq 1$ and $1/M = 2^{-(r+1)}$, which proves the last assertion. ◀

233 ► **Lemma 4.4** (Exact homogeneous output). *Whenever the algorithm reaches its final
234 verified output, that output is a K -clique or a K -vertex independent set.*

235 **Proof.** For $i < j$, the definition of S_{i+1} gives $G(v_i, v_j) = c_i$, and it also gives $G(v_i, w) =$
236 c_i . Among the $r = 2K - 3$ colours, one occurs at least $\lceil r/2 \rceil = K - 1$ times. The
237 corresponding pivots together with w therefore induce that colour. The final direct
238 check verifies the returned set independently of earlier search or estimation errors. ◀

239 **5 Query complexity and a classical lower bound**

240 For the size-biased recursion, a capped sample from a nonempty subset of the M -
241 element universe costs $O(\sqrt{M})$ membership tests. Repeating the block $O(\log(2r))$
242 times gives failure probability $O(1/r)$. Since the membership predicate at level i costs
243 $O(i + 1)$ edge queries, one fixed-cap run uses

$$244 \quad O\left(\sqrt{M} \log(2r) \sum_{i=0}^{r-1} (i + 1)\right) = O(K^2 2^K \log K)$$

245 edge queries. The amplification in Section 4 gives the following worst-case bound.

► **Proposition 5.1** (Size-biased quantum recursion). *Under the hypotheses of Theorem 1.1, the estimation-free algorithm returns a verified homogeneous K -set with probability at least $1 - \eta$ using*

$$O\left(K^2 2^K \log K \log \frac{1}{\eta}\right)$$

edge queries.

We now analyze the sharper scale-aware recursion. By Lemma 4.3, a capped search block at level i uses $O(2^{i/2})$ membership tests. Computing and uncomputing membership and querying the sampled edge costs $O(i + 1)$ edge queries. Define

$$b_i = (i + 1)2^{i/2}.$$

The batch cap uses $O(m_i + \log(r/\eta)) = O(m_i)$ blocks, so the total sampling cost is

$$O\left(\log \frac{r}{\eta} \sum_{i=0}^{r-1} b_i \varepsilon_i^{-2}\right). \quad (5)$$

At most six indices share any value of d_i , and hence

$$Z = \sum_i 2^{-d_i} < 6 \sum_{d \geq 0} 2^{-d} = 12.$$

Using Equation (1),

$$\sum_{i=0}^{r-1} b_i \varepsilon_i^{-2} = O\left(\sum_{i=0}^{r-1} (i + 1)2^{i/2} 4^{d_i}\right).$$

Write $h = r - 1 - i$. Since $4^{\lceil h/6 \rceil} \leq 4 \cdot 2^{h/3}$, the last display is at most

$$O\left(2^{r/2} \sum_{h=0}^{r-1} (r - h)2^{-h/6}\right) = O(r2^{r/2}).$$

The high-confidence pivot searches, the final search, and the K^2 output verification satisfy the same bound. Substituting $r = 2K - 3$ into Equation (5) proves Theorem 1.1.

Each search begins in the uniform state on the fixed power-of-two universe $[M]$. The recorded pivot labels and colours define the reversible membership predicate for S_i , so the algorithm requires no materialized representation of any candidate set.

We complement the upper bound with a randomized classical lower bound. The following proposition follows from the random-Painter analysis in [4]; Appendix B gives the derivation.

► **Proposition 5.2** (Randomized classical lower bound). *For $K \geq 8$ and $M = 4^{K-1}$, every randomized edge-query algorithm that finds a homogeneous K -set with worst-case probability at least $2/3$ uses*

$$\Omega\left(2^{(2-\sqrt{2})K}\right) = \Omega\left(M^{1-1/\sqrt{2}}\right)$$

queries.

The exponent $1 - 1/\sqrt{2} \approx 0.292893$ improves the earlier $M^{1/4}$ black-box lower bound [9, 11].

The size-biased recursion also extends to every fixed number of edge colours. The statement and recurrence appear in Appendix C.1.

6 Related work and a reported lower-bound discrepancy

Several quantum approaches to Ramsey problems address a different input and output task from the one studied here. Work based on adiabatic computation, quantum annealing, or quantum counting searches for colourings that establish or test Ramsey-number bounds [7, 1, 21, 19, 20, 18]. In those formulations, the search space consists of labelled graph colourings. Our input is instead one fixed graph given by an adjacency oracle, and the goal is to find a homogeneous set guaranteed to exist in that graph.

Generic quantum algorithms for clique, independent set, and fixed-subgraph containment operate in a closer oracle model [5, 3, 14, 22]. Quantum backtracking supplies another general search primitive [16]. The present algorithm instead exploits the nested majority structure of the constructive Ramsey proof by representing each candidate set through the adjacency constraints accumulated along the recursion and sampling directly from the resulting implicit set. We are not aware of an earlier implementation of the constructive Erdős–Szekeres recursion by this mechanism in the adjacency-query model.

Ramsey search also appears in proof complexity and TFNP. It is connected to weak and structured pigeonhole principles [12, 13], and belongs to the polynomial long-choice class PLC [17]. The closest matching query relation is studied in [10].

Alignment of the formal parameters.

Definition II.6 of [10] uses $N = 2^n$ vertices, and the convention following Definition II.7 sets the default target to $K = n/2$. The unnumbered “Query complexity of RAMSEY” paragraph in Section III (p. 415) reports an $N^{1-o(1)}$ quantum lower bound, based on its Lemma III.1 and the multicollision lower bound of [15].

The graph produced by Lemma III.1 can be written, away from the diagonal, as

$$A_h(u, v) = \mathbf{1}[h(u) = h(v)] \vee A_0(h(u), h(v)), \quad A_h(u, u) = 0.$$

This is a symmetric simple graph, and a coherent query to it uses $O(1)$ queries to the value oracle for h , including uncomputation of the two hash values. Thus the reduction lies within our valid-graph promise. Corollary 2.1 gives the same upper bound for the locally verifiable arbitrary-circuit relation. Moreover, Corollary 1.2 returns $\lfloor n/2 \rfloor + 1$ vertices, from which one may discard a vertex to meet the default target. These observations align the graph promise, encoding, and output size in the two formulations.

312 Direct parameter substitution.

313 Let H denote the multicollision range size and G the number of vertices in the resulting
 314 Ramsey instance. For fixed multiplicity t , the sufficient condition in Theorem I.3 of
 315 [10] is $G \geq H^{4t}/4^t$. At the boundary $G = \Theta_t(H^{4t})$, the exponent in [15] is

$$316 \quad \alpha_t = \frac{2^{t-1} - 1}{2^t - 1}.$$

317 The exponent transferred to G is therefore

$$318 \quad \frac{\alpha_t}{4t} = \frac{2^{t-1} - 1}{4t(2^t - 1)} \leq \frac{1}{24}, \quad t \geq 2.$$

319 Equality holds at $t = 2$. For $t \geq 3$, the inequality $\alpha_t < 1/2$ gives $\alpha_t/(4t) < 1/(8t) \leq$
 320 $1/24$. Increasing G relative to H can only reduce the transferred exponent. The
 321 multicollision theorem is stated for fixed t , so this substitution also does not support
 322 a growing-multiplicity limit.

323 Thus direct composition of the two cited statements yields an exponent at most $1/24$,
 324 rather than $1 - o(1)$. If the reported near-linear consequence uses the same bounded-
 325 error query relation and parameter convention, it is incompatible with Corollary 1.2.
 326 An additional ingredient or a different normalization may account for the discrepancy.

327 7 Discussion

328 The query improvement comes from combining two ideas. First, the algorithm stores
 329 each candidate set as a conjunction of the adjacency constraints created by earlier
 330 pivots. This representation supports quantum sampling without enumerating the
 331 surviving vertices. Second, the scale-aware analysis allocates estimation error according
 332 to the cost of the recursion level. The resulting schedule keeps the cumulative loss in
 333 candidate-set size bounded while concentrating most of the sampling effort where it is
 334 cheapest.

335 A natural next step is to determine the randomized classical query complexity in
 336 the range between Proposition 5.2 and Theorem 1.1. The implicit-set method may
 337 also extend to off-diagonal or hypergraph Ramsey search. Quantum walks already
 338 give sublinear-query algorithms for fixed sub-hypergraph containment [8]; a Ramsey
 339 algorithm would additionally need to exploit the totality and recursive structure of
 340 the problem. For multiple edge colours, Corollary C.1 gives a size-biased algorithm,
 341 while a scale-aware analogue could yield a sharper dependence on the recursion depth.

342 Appendix C.2 reports finite experiments that illustrate the survival recurrence and
 343 the conditional uniformity of the sampler. The main query bound follows from the
 344 analytic argument in Sections 2–5.

345 The constructive diagonal Ramsey recursion admits a quantum implementation
 346 over implicit candidate sets. Capped uniform search and a scale-aware concentration
 347 schedule give $O(2^K K \log(K/\eta))$ edge queries, or $\tilde{O}(\sqrt{N})$ queries at the succinct Ramsey
 348 scale.

A Capped search and batch uniformity

We prove the properties of the search primitive used in Lemma 2.2. Begin in the uniform superposition on $[M]$ and apply the randomized-iteration unknown-solution search of [2]. When the marked density is at least λ , truncate the geometric iteration schedule once its maximum iteration count reaches a sufficiently large constant multiple of $1/\sqrt{\lambda}$. The standard analysis gives a success probability $p_0 > 0$, independent of M and λ , using $O(1/\sqrt{\lambda})$ calls to the membership predicate. Because the cap is fixed before any oracle outcome is observed, the block terminates for every predicate.

For each fixed number of Grover iterations, all marked basis states have the same amplitude. Randomizing the iteration count preserves this permutation symmetry. After measuring a candidate, evaluate the exact membership predicate and accept only a marked outcome. Conditional on acceptance, the output is therefore uniform on the marked set T . Independent blocks reinitialize all registers and random choices, so their accepted outputs are independent conditional on the block-success indicators.

To collect a batch, run

$$B = C(m + \log(1/\delta))$$

independent capped blocks, where the absolute constant C is chosen from p_0 . The number of accepted blocks stochastically dominates $\text{Bin}(B, p_0)$. A Chernoff lower-tail bound makes the probability of fewer than m acceptances at most δ . Conditional on any success pattern with at least m acceptances, the first m accepted outputs are independent and uniform on T . This proves Lemma 2.2.

B Derivation of the randomized classical lower bound

We derive Proposition 5.2 from the random-Painter weight bound in [4]. Let

$$\beta = 2 - \sqrt{2}, \quad \alpha = 1 - \frac{1}{\sqrt{2}},$$

so that $\beta = 2\alpha$. Set

$$T = \lfloor 2^{\beta K - 2} \rfloor, \quad c = \lfloor \alpha K \rfloor = \alpha K + \epsilon, \quad -1 < \epsilon \leq 0.$$

Expose $G(M, 1/2)$ adaptively to a deterministic algorithm making T edge queries. The cited random-Painter estimate bounds the probability that the queried edges contain a fully exposed monochromatic K -set by

$$2 \cdot 2^{-\binom{K}{2} + c(c-1)} (2T)^{K-c}. \tag{6}$$

Since $2T \leq 2^{\beta K - 1}$, the base-2 logarithm of Equation (6) is at most

$$1 - \binom{K}{2} + c(c-1) + (K-c)(\beta K - 1) = 1 - \frac{K}{2} + \epsilon^2.$$

For $K \geq 8$, this quantity is at most -2 , so the event has probability at most $1/4$.

Now condition on the query transcript and suppose that the algorithm outputs a K -set whose internal edges have not already been fully exposed in one colour. If the exposed internal edges contain both colours, the output is not homogeneous. If they contain exactly one colour, at least one unexposed fair edge must match that colour, which occurs with conditional probability at most $1/2$. If no internal edge has been exposed, the success probability is $2^{1-\binom{K}{2}} \leq 1/2$ for $K \geq 3$. Thus the conditional success probability is at most $1/2$ outside the fully exposed event, and the total success probability under $G(M, 1/2)$ is at most

$$\frac{1}{4} + \frac{3}{4} \cdot \frac{1}{2} = \frac{5}{8} < \frac{2}{3}.$$

Yao's minimax principle implies that a randomized algorithm with worst-case success probability at least $2/3$ must make more than T queries. Finally, $M = 4^{K-1}$ gives

$$2^{\beta K} = \Theta(M^{\beta/2}) = \Theta(M^{1-1/\sqrt{2}}),$$

which proves Proposition 5.2.

C Multicolour extension and numerical experiments

C.1 A multicolour extension of size-biased survival

The size-biased mechanism extends to any number of colours. Suppose every edge has one of $q \geq 2$ colours, and define

$$r = q(K-2) + 1, \quad B_d = 1 + q + \dots + q^{d-1} = \frac{q^d - 1}{q - 1}.$$

Choose the power of two

$$L_q = 2^{\lceil \log_2(2B_r) \rceil},$$

so $2B_r \leq L_q < 4B_r$. If the $s-1$ non-pivot vertices split into classes of sizes $a_1, \dots, a_q$, the induction in Lemma 4.1 becomes

$$\sum_{j=1}^q (a_j - B_{d-1})_+ \geq (s-1 - qB_{d-1})_+ = (s - B_d)_+,$$

because $B_d = 1 + qB_{d-1}$. Thus, for $d \geq 1$, the d -step survival probability is at least $(s - B_d)_+ / (s-1)$ when $s \geq 2$. Starting from $L_q \geq 2B_r$ gives survival probability greater than $1/2$. Among the r recorded colours, at least one occurs $K-1$ times.

► **Corollary C.1** (Multicolour size-biased Ramsey search). *For $q, K \geq 2$, every q -edge-coloured complete graph on at least L_q vertices admits a bounded-error quantum algorithm that outputs a monochromatic K -clique using*

$$O\left(r^2 \sqrt{L_q} \log(2r) \log \frac{1}{\eta}\right)$$

colour-oracle queries in the worst case, where r , B_r , and L_q are as defined above.

C.2 Small-instance numerical experiments

Ideal-sampler recursion. We ran the scale-aware recursion at $K = 3$, $N = 16$, and $r = 3$ for 1,000 trials in each of five graph families. The error schedule was $(\varepsilon_0, \varepsilon_1, \varepsilon_2) = (1/64, 1/64, 1/32)$, with sample counts $m_i = \lceil \log(600)/(2\varepsilon_i^2) \rceil$, corresponding to a total estimation-failure budget of 0.01. Pivots were sampled uniformly, majority-estimation samples were independent with replacement, and ties selected colour 1. We replaced quantum search by ideal uniform sampling to isolate the combinatorial recursion.

The deterministic inputs were the complete graph, the empty graph, and the balanced complete bipartite graph $K_{8,8}$. For the random inputs, one $G(16, 1/2)$ graph was held fixed across all 1,000 trials, while the final family drew a fresh $G(16, 1/2)$ graph in each trial. Every returned triple was verified; exhaustion or a nonhomogeneous output counted as failure.

Graph family on 16 vertices	failed/trials	minimum final $ S_3 $
complete	0/1000	13
empty	0/1000	13
balanced complete bipartite $K_{8,8}$	0/1000	6
one fixed $G(16, 1/2)$	0/1000	2
fresh $G(16, 1/2)$ per trial	0/1000	2

All 5,000 trials returned homogeneous triples. The final column records the smallest observed value of $|S_3|$ in each family.

Isolated Grover sampler. We evolved a 16-dimensional state vector initialized in the uniform superposition, using marked-set sizes 1, 2, 3, 4, 8, 15. The measurement distribution was averaged over a uniformly chosen number of Grover iterations from zero through four. The probability of measuring a marked item ranged from 0.4026746749 to 0.5973253251. Conditional on a marked outcome, the largest deviation from the uniform distribution on marked items was zero to machine precision. This calculation isolates the permutation symmetry used in Lemma 2.2.

Finite split-tree survival. Minimizing the ideal size-biased survival probability over every admissible colour-class split gives exact finite-depth comparisons with the analytic bounds. For nine binary levels starting from 1,024 candidates, the minimum was approximately 0.5086135566, above the lower bound 513/1024. For four ternary levels starting from 128 candidates, the minimum was approximately 0.7155732457, above 88/127.

C.3 AI-assisted research workflow

ChatGPT and Codex were used to brainstorm candidate recurrences and proof strategies, critique proof drafts, assist with the numerical scripts, and edit the manuscript. The authors reconstructed and checked every argument, implementation, numerical value, and reference. The formal disclosure appears at the end of the paper.

References

- 1 Zhengbing Bian, Fabian Chudak, William G. Macready, Geordie Rose, and Frank Gaitan. Experimental determination of Ramsey numbers. *Physical Review Letters*, 111:130505, 2013. URL: <https://arxiv.org/abs/1201.1842>, arXiv:1201.1842, doi:10.1103/PhysRevLett.111.130505.
- 2 Michel Boyer, Gilles Brassard, Peter Høyer, and Alain Tapp. Tight bounds on quantum searching. *Fortschritte der Physik*, 46(4–5):493–506, 1998. URL: <https://arxiv.org/abs/quant-ph/9605034>, arXiv:quant-ph/9605034.
- 3 Andrew M. Childs and Jason M. Eisenberg. Quantum algorithms for subset finding. *Quantum Information and Computation*, 5(7):593–604, 2005. arXiv:quant-ph/0311038, doi:10.26421/QIC5.7–7.
- 4 David Conlon, Jacob Fox, Andrey Grinshpun, and Xiaoyu He. Online Ramsey numbers and the subgraph query problem. In *Building Bridges II: Mathematics of László Lovász*, volume 28 of *Bolyai Society Mathematical Studies*, pages 159–194. Springer, 2020. URL: <https://arxiv.org/abs/1806.09726>, arXiv:1806.09726.
- 5 Sebastian Dörn. Quantum complexity bounds for independent set problems. *arXiv preprint*, 2005. URL: <https://arxiv.org/abs/quant-ph/0510084>, arXiv:quant-ph/0510084.
- 6 Paul Erdős and George Szekeres. A combinatorial problem in geometry. *Compositio Mathematica*, 2:463–470, 1935. URL: <https://eudml.org/doc/88611>.
- 7 Frank Gaitan and Lane Clark. Ramsey numbers and adiabatic quantum computing. *Physical Review Letters*, 108:010501, 2012. URL: <https://arxiv.org/abs/1103.1345>, arXiv:1103.1345.
- 8 François Le Gall, Harumichi Nishimura, and Seiichiro Tani. Quantum algorithms for finding constant-sized sub-hypergraphs. *Theoretical Computer Science*, 609:569–582, 2016. arXiv:1310.4127, doi:10.1016/j.tcs.2015.10.006.
- 9 Russell Impagliazzo and Moni Naor. Decision trees and downward closures. In *Proceedings of the Third Annual Structure in Complexity Theory Conference*, pages 29–38. IEEE, 1988. doi:10.1109/SCT.1988.5260.
- 10 Siddhartha Jain, Jiawei Li, Robert Robere, and Zhiyang Xun. On pigeonhole principles and Ramsey in TFNP. In *2024 IEEE 65th Annual Symposium on Foundations of Computer Science (FOCS)*, pages 406–428. IEEE, 2024. arXiv:2401.12604, doi:10.1109/FOCS61266.2024.00033.
- 11 Ilan Komargodski, Moni Naor, and Eylon Yogev. White-box vs. black-box complexity of search problems: Ramsey and graph property testing. *Journal of the ACM*, 66(5):34:1–34:28, 2019. doi:10.1145/3341106.
- 12 Jan Krajíček. On the weak pigeonhole principle. *Fundamenta Mathematicae*, 170(1–2):123–140, 2001. doi:10.4064/fm170-1-8.
- 13 Jan Krajíček. Structured pigeonhole principle, search problems and hard tautologies. *The Journal of Symbolic Logic*, 70(2):619–630, 2005. doi:10.2178/jsl/1120224731.
- 14 Troy Lee, Frédéric Magniez, and Miklos Santha. A learning graph based quantum query algorithm for finding constant-size subgraphs. *Chicago Journal of*

- 491 *Theoretical Computer Science*, 2012(10), 2012. URL: [https://arxiv.org/abs/](https://arxiv.org/abs/1109.5135)
 492 1109.5135, arXiv:1109.5135.
- 493 **15** Qipeng Liu and Mark Zhandry. On finding quantum multi-collisions. In *Advances*
 494 *in Cryptology—EUROCRYPT 2019, Part III*, volume 11478 of *Lecture Notes*
 495 *in Computer Science*, pages 189–218. Springer, 2019. arXiv:1811.05385,
 496 doi:10.1007/978-3-030-17659-4_7.
- 497 **16** Ashley Montanaro. Quantum walk speedup of backtracking algorithms. *Theory*
 498 *of Computing*, 14(15):1–24, 2018. URL: <https://arxiv.org/abs/1509.02374>,
 499 arXiv:1509.02374, doi:10.4086/toc.2018.v014a015.
- 500 **17** Amol Pasarkar, Christos Papadimitriou, and Mihalis Yannakakis. Extremal
 501 combinatorics, iterated pigeonhole arguments and generalizations of PPP. In
 502 *14th Innovations in Theoretical Computer Science Conference (ITCS 2023)*,
 503 volume 251 of *Leibniz International Proceedings in Informatics (LIPIcs)*, pages
 504 88:1–88:20. Schloss Dagstuhl–Leibniz-Zentrum für Informatik, 2023. doi:10.4
 505 230/LIPIcs.ITCS.2023.88.
- 506 **18** Joel E. Pion and Susan M. Mniszewski. Toward computing bounds for Ramsey
 507 numbers using quantum annealing. *Quantum Information Processing*, 24(7):221,
 508 2025. URL: <https://arxiv.org/abs/2311.04405>, arXiv:2311.04405,
 509 doi:10.1007/s11128-025-04839-x.
- 510 **19** Ri Qu, Zong shang Li, Juan Wang, Yan ru Bao, and Xiao chun Cao. Computing
 511 hypergraph Ramsey numbers by using quantum circuit. *Quantum Information*
 512 *Processing*, 12(7):2487–2496, 2013. arXiv:1210.3419, doi:10.1007/s11128-0
 513 13-0541-9.
- 514 **20** Mani Ranjbar, William G. Macready, Lane Clark, and Frank Gaitan. Generalized
 515 Ramsey numbers through adiabatic quantum optimization. *Quantum Information*
 516 *Processing*, 15(9):3519–3542, 2016. arXiv:1606.01078, doi:10.1007/s11128-0
 517 16-1363-3.
- 518 **21** Hefeng Wang. Determining Ramsey numbers on a quantum computer. *Physical*
 519 *Review A*, 93(3):032301, 2016. arXiv:1510.01884, doi:10.1103/PhysRevA.93.
 520 032301.
- 521 **22** Yechao Zhu. Quantum query complexity of constant-sized subgraph containment.
 522 *International Journal of Quantum Information*, 10(3):1250019, 2012. arXiv:
 523 1109.4165, doi:10.1142/S0219749912500190.

524 **AI Disclosure.** We used ChatGPT and Codex to assist with brainstorming candidate
 525 algorithms and proof strategies, reviewing numerical experiments, and editing the
 526 manuscript. These tools materially influenced Sections 1–7 and Appendix C; their role
 527 in the research workflow is described in Appendix C.3. The authors independently
 528 verified the correctness and originality of all content, including every theorem, proof,
 529 calculation, experimental result, and reference, and take full responsibility for the
 530 paper.